Solution 1

Problem 1 — For $x \in [0,1)$, set $\tau(x) = 2x \mod 1$. Show that τ is a measure-preserving transformation of $([0,1), \mathscr{B}([0,1)), dx)$, and that τ is ergodic. Identify the invariant function $\overline{f}$ corresponding to each integrable function f.

Alright, let's recall the definitions.

- A **measure-preserving transformation** τ satisfies
$$\mu \circ \tau^{-1} = \mu$$
over $\mathscr{B}$.
- An **invariant subset** $A \in \mathscr{B}$ satisfies $\tau^{-1}(A) = A$; these form a sigma algebra $\mathscr{B}_\tau$.
- If $\mathscr{B}_\tau$ contains only sets of measure 0 or 1, then τ is **ergodic**.

First off, for all $0 \le a < b < 1$,
$$\begin{aligned}\mu\left(\tau^{-1}([a,b))\right) &= \mu\left(\left[\frac{a}{2}, \frac{b}{2}\right)\right) + \mu\left(\left[\frac{a+1}{2}, \frac{b+1}{2}\right)\right) \\ &= \frac{b-a}{2} + \frac{b-a}{2} = b - a = \mu([a,b)).\end{aligned}$$
Recall that $\mathscr{B}([0,1))$ is generated by the intervals $[a,b)$, thus τ is measure-preserving.

Next, consider any invariant subset $\tau^{-1}(A) = A$. Let
$$c_k = \frac{1}{2\pi}\int_0^1 \mathbb{1}_A e^{-2\pi i k x} dx, \qquad k \in \mathbb{Z}$$
be the **Fourier series** of $\mathbb{1}_A$. Then,
$$\begin{aligned}c_k &= \frac{1}{2\pi}\int_0^1 \mathbb{1}_A e^{-2\pi i k x} dx \\ &= \frac{1}{2\pi}\int_0^1 \left(\mathbb{1}_A e^{-2\pi i k x}\right) \circ \tau dx \\ &= \frac{1}{2\pi}\int_0^1 \mathbb{1}_{\tau^{-1}(A)} e^{-2\pi i k \tau(x)} dx \\ &= \frac{1}{2\pi}\int_0^1 \mathbb{1}_A e^{-2\pi i k 2x} dx \\ &= c_{2k}.\end{aligned}$$
Thus, $c_k = c_{2k} = c_{4k} = \cdots$ for all $k \in \mathbb{Z}$. By Parseval's identity, we know
$$\sum_{k\in\mathbb{Z}} |c_k|^2 = \frac{1}{(2\pi)^2}\int_0^1 |\mathbb{1}_A|^2 dx = \frac{1}{(2\pi)^2}\mu(A)$$
Now, the RHS is supposed to be finite. Thus, all c_k for $k \neq 0$ must be 0. Thus, $\mathbb{1}_A$ must be constant almost everywhere (by considering $\mathbb{1}_A - 2\pi c_0$). In other words, A must have measure 0 or 1; thus, τ is ergodic.

Finally, we want to identify the invariant function $\overline{f}$ corresponding to an integrable function f.

> **Idea —** This is obviously just $\mathbb{E}[f]$; I guess you want me to show that this actually works. Let's blow it out the water with the like, 17 theorems we know about this.

I interpret this to mean the almost-sure limit of S_n/n guaranteed by Birkhoff's ergodic theorem, which by von Neumann's L^1 ergodic theorem is also the L^1 limit of $\frac{S_n}{n}$. I already know a priori that you can take $\overline{f}$ to be constant, because invariant functions on ergodic systems are constant almost everywhere. So let $\overline{f} = c$ or something. Because this is the L^1 limit, I know that

$$\int_{[0,1)} \left| \frac{S_n}{n} - c \right| \geq \left| \int_{[0,1)} \frac{S_n}{n} - c \right|$$

but the LHS converges to 0. Thus c is the limit of

$$\int_{[0,1)} \frac{S_n}{n} = \frac{1}{n} \sum_{k=1}^{n} \int_{[0,1)} f \circ \tau^k = \frac{1}{n} \sum_{k=1}^{n} \int_{[0,1)} f = \frac{1}{n} \sum_{k=1}^{n} \mathbb{E}[f] = \mathbb{E}[f].$$

Solution 2

Problem 2 — Fix $a \in [0,1)$ and define, for $x \in [0,1)$, $\tau(x) = x + a \mod 1$. Show that τ is also a measure-preserving transformation of $([0,1), \mathscr{B}([0,1)), dx)$. Determine for which values of a the transformation τ is ergodic. Identify, for all values of a, the invariant function f corresponding to an integrable function f.

τ is measure-preserving because uh

$$\begin{aligned}\mu(\tau^{-1}(E)) &= \mu(\tau^{-1}(E \cap [0, 1-a))) + \mu(\tau^{-1}(E \cap [1-a, 1))) \\ &= \mu(E \cap [0, 1-a)) + \mu(E \cap [1-a, 1)) = \mu(E).\end{aligned}$$

where the second equality is because the Lebesgue measure is defined to be translation-invariant.

Obviously τ is ergodic iff a is irrational; let me prove it.

𝔖2.1 a is rational

If $\tau = p/q$ when reduced where $p, q \in \mathbb{Z}$, then the set

$$\left(\frac{m}{q}, \frac{m+1/2}{q}\right), \qquad 0 \le m \le q-1$$

is invariant, and has measure $1/2$. In this case, the invariant function can be explicitly constructed:

$$\frac{S_{nq}}{nq}(x) = \frac{1}{nq}\sum_{k=1}^{nq} f \circ \tau^k(x) = \frac{1}{nq}\sum_{k=1}^{nq} f\left(x + \frac{kp}{q}\right) = \frac{1}{q}\sum_{k=0}^{q-1} f\left(x + \frac{k}{q}\right)$$

𝔖2.2 a is irrational

Now, pick some invariant A and define the Fourier series of $\mathbb{1}_A$ as before. Then,

$$\begin{aligned}c_k &= \frac{1}{2\pi}\int_0^1 \mathbb{1}_A e^{-2\pi i k x} dx \\ &= \frac{1}{2\pi}\int_0^1 \left(\mathbb{1}_A e^{-2\pi i k x}\right) \circ \tau dx \\ &= \frac{1}{2\pi}\int_0^1 \mathbb{1}_{\tau^{-1}(A)} e^{-2\pi i k \tau(x)} dx \\ &= \frac{1}{2\pi}\int_0^1 \mathbb{1}_A e^{-2\pi i k (x+a)} dx \\ &= \frac{1}{2\pi}\int_0^1 \mathbb{1}_A e^{-2\pi i k x} dx e^{-2\pi i k a} \\ &= c_k e^{-2\pi i k a}.\end{aligned}$$

a is irrational, thus $e^{-2\pi i k a} \neq 1$ for all $k \neq 0$, thus $c_k = 0$ for all $k \neq 0$, and then f is constant almost everywhere, thus A has measure 0 or 1. Thus τ is ergodic.

Finally, we want to identify the invariant function. By identical reasoning to problem 1, $\overline{f}$ must be constant almost everywhere, and $\overline{f} = \mathbb{E}[f]$.

Solution 3

Problem 3 — Call a sequence of random variables $(X_n)_{n\in\mathbb{N}}$ on a probability space stationary if for each $n, k \in \mathbb{N}$ the random vectors $(X_1, \ldots, X_n)$ and $(X_{k+1}, \ldots, X_{k+n})$ have the same distribution: for $A_1, \ldots, A_n \in \mathscr{B}$,

$$\mathbb{P}[X_1 \in A_1, \ldots, X_n \in A_n] = \mathbb{P}[X_{k+1} \in A_1, \ldots, X_{k+n} \in A_n].$$

Show that, if $(X_n)_{n\in\mathbb{N}}$ is a stationary sequence and $X_1 \in L^p$, for some $p \in [1, \infty)$, then

$$\frac{1}{n}\sum_{i=1}^{n} X_i \to X \text{ a.s. and in } L^p,$$

for some random variable $X \in L^p$ and find $\mathbb{E}[X]$.

Okay, I can push the measure forward by considering the map $X : \Omega \to \mathbb{R}^{\mathbb{N}}$ via $\omega \mapsto (X_1(\omega), X_2(\omega), \ldots)$; here, $\mathbb{R}^{\mathbb{N}}$ is equipped with the product sigma algebra (generated by the cylinder sets with finitely many coordinates fixed); then let $\mu = \mathbb{P} \circ X^{-1}$.

Consider the shift map $\theta : \Omega \to \Omega$ defined by $\theta(\omega) = (\omega_2, \omega_3, \ldots)$. Then, the condition precisely states that θ is measure-preserving; it preserves the measure on all cylinder sets, and these generate the product sigma algebra.

Thus, von Neumann's L^p ergodic theorem applied to the coordinate function $X_1 \circ X^{-1} : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}$ shows $\frac{S_n}{n} \to X$ AS and in L^p, where $S_n = X_1 + \cdots + X_n$, for some rv X.

Note that $L^p \subset L^1$ for all $p \in [1, \infty)$ so $X_1 \in L^1$ as well, and $\frac{S_n}{n} \to X$ in L^1 as well.

Observe that

$$\mathbb{E}\left[\frac{X_1 + \cdots + X_n}{n}\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] = \mathbb{E}[X_1].$$

for all n. Also,

$$\left|\mathbb{E}[X] - \mathbb{E}\left[\frac{S_n}{n}\right]\right| \leq \int_\Omega \left|X - \frac{S_n}{n}\right| d\mu \to 0.$$

Thus, $\mathbb{E}[X] = \mathbb{E}[X_1]$.

Solution 4

Problem 4 — Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent random variables, such that $\mathbb{E}[X_n] = \mu$ and $\mathbb{E}[X_n^4] \leq M$ for all n, for some constants $\mu \in \mathbb{R}$ and $M < \infty$. Set $P_n = X_1X_2 + X_2X_3 + \cdots + X_{n-1}X_n$. Show that P_n/n converges a.s. as $n \to \infty$ and identify the limit.

We claim $\frac{P_n}{n} \to \mu^2$ almost surely.

The trick is to split the sequence into even and odd components;

$$P_n = (X_1X_2 + X_3X_4 + \dots) + (X_2X_3 + X_4X_5 + \dots)$$

Each of these subsequences contains independent terms. The mean of each term is $\mathbb{E}[X_iX_{i+1}] = \mathbb{E}[X_i]\mathbb{E}[X_{i+1}] = \mu^2$, and the fourth moment is bounded by M^2 by independence.

Thus, theorem 10.1.1 says

$$\frac{X_1X_2 + \cdots + X_{n-1}X_n}{n/2} \to \mu^2$$

almost surely; we are ignoring off-by-one errors due to the parity of n because those do not matter in the limit. Thus, the average of the entire sum converges a.s. to μ^2 as well.

Solution 5

Problem 5.1 — The Cauchy distribution has density function $f(x) = \frac{1}{\pi}\frac{1}{1+x^2}$ for $x \in \mathbb{R}$. Show that the corresponding characteristic function is given by $\phi(u) = e^{-|u|}$.

Okay, let us compute

$$\phi(u) = \mathbb{E}[e^{iuX}] = \int_{-\infty}^{\infty} e^{iux}\frac{1}{\pi}\frac{1}{1+x^2}dx.$$

This is clearly symmetric about $u \to -u$ by sending $x \to -x$. Thus we consider the case $u \geq 0$ only.

We can compute this by contour integration. Consider the contour γ composed of the semi-circle γ_R of radius R in the upper half-plane centered at 0, and the line segment $[-R, R]$ on the real axis. In the limit, the integral over the semi-circle is bounded by

$$\begin{aligned}
\left|\int_{\gamma_R} e^{iuz}\frac{1}{\pi}\frac{1}{1+z^2}dz\right| &\leq \pi R \cdot \sup_{z\in\gamma_R}\left|e^{iuz}\frac{1}{\pi}\frac{1}{1+z^2}\right| \\
&\leq \pi R \cdot \sup_{z\in\gamma_R}\underbrace{\left|e^{-u\mathrm{Im}(z)}\right|_{<1}\frac{1}{\pi}\frac{1}{1+z^2}} \\
&\leq \pi R \cdot \frac{1}{\pi}\frac{1}{R^2-1} \\
&\to 0
\end{aligned}$$

thus the whole contour integral becomes the integral over the real line.

On the other hand, by the residue theorem, the integral over the contour is $2\pi i$ times the sum of the residues of the poles inside the contour. The poles are at $z = \pm i$, and the residue at $z = i$ is

$$\lim_{z\to i}(z-i)e^{iuz}\frac{1}{\pi}\frac{1}{1+z^2} = \frac{e^{-u}}{2i}$$

Thus, the integral is $2\pi i \cdot \frac{e^{-u}}{2i} = \pi e^{-u}$. Thus, $\phi(u) = e^{-|u|}$, as desired.

Problem 5.2 — Show also that, if $X_1, \ldots, X_n$ are independent Cauchy random variables, then the random variable $Y = (X_1 + \cdots + X_n)/n$ is also Cauchy.

We compute the characteristic function of Y:

$$\phi_Y(u) = \mathbb{E}[e^{iuY}] = \mathbb{E}[e^{iu(X_1+\cdots+X_n)/n}] = \mathbb{E}[e^{iuX_1/n}]^n = \phi(u/n)^n = e^{-|u|}.$$

Thus, Y is Cauchy.

Solution 6

Problem 6.1 — For each $n \in \mathbb{N}$, there is a unique probability measure on the sphere $S^{n-1} = \{x \in \mathbb{R}^n : |x| = 1\}$ such that $\mu_n(A) = \mu_n(UA)$ for all Borel sets A and all orthogonal $n \times n$ matrices U. Fix $k \in \mathbb{N}$ and, for $n \geq k$, let γ_n denote the probability measure on $\mathbb{R}^k$ which is the law of $\sqrt{n}(x_1, \ldots, x_k)$ under μ_n.

Show that if X is a mean-zero Gaussian with covariance matrix given by I_n then $X/|X|$ has law given by μ_n.

Observe that the density function (and law) of X is in fact symmetric under orthogonal transformations;

$$\mathcal{N}(0, I_n)(Ux) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}(Ux)^T Ux} = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}x^T x} = \mathcal{N}(0, I_n)(x).$$

Thus, the law of $X/|X|$ is also symmetric under orthogonal transformations, and is a probability measure on the sphere S^{n-1}. Thus, it must be μ_n.

Problem 6.2 — Show that if $(X_n)_{n\in\mathbb{N}}$ is a sequence of independent $N(0,1)$ random variables and if $R_n = \sqrt{X_1^2 + \cdots + X_n^2}$ then $R_n/\sqrt{n} \to 1$ a.s.

The sequence $(X_n^2)_{n\in\mathbb{N}}$ is a sequence of i.i.d random variables with mean 1 (this is just the χ^2 distribution).

Thus, by the strong law of large numbers, $\frac{X_1^2+\cdots+X_n^2}{n} \to 1$ almost surely. Taking the square root of each side, $\frac{R_n}{\sqrt{n}} \to 1$ almost surely.

Problem 6.3 — Show that γ_n converges weakly to the standard Gaussian distribution on $\mathbb{R}^k$ as $n \to \infty$.

By part 1,

$$\gamma_n \sim \sqrt{n}\frac{X_k}{|X_n|}$$

where $X_i = (x_1, \ldots x_i)$ and each $(x_i)_{i\geq n}$ is a standard Gaussian.

By the previous problem, it is also true that

$$\frac{|X_n|}{\sqrt{n}} \to 1 \text{ almost surely.}$$

so it also converges in probability. Thus, by Slutsky's theorem, $\gamma_n \to \mathcal{N}(0, I_k)$ in distribution.

Solution 7

Problem 7 — Let X and Y be integrable random variables and suppose that $\mathbb{E}[X|Y] = Y$ and $\mathbb{E}[Y|X] = X$ a.s. Show that $X = Y$ a.s. Hint: Consider quantities like $\mathbb{E}[(X-Y)\mathbb{1}_{X>c,Y\le c}] + \mathbb{E}[(X-Y)\mathbb{1}_{X\le c,Y\le c}]$.

Well, by definition of expected value, $\mathbb{E}[Y\mathbb{1}_{Y\in A}] = \mathbb{E}[X\mathbb{1}_{Y\in A}]$ for all A and $\mathbb{E}[X\mathbb{1}_{X\in A}] = \mathbb{E}[Y\mathbb{1}_{X\in A}]$ for all A. For all $c \in \mathbb{R}$,

$$\begin{aligned}\mathbb{E}[(X-Y)\mathbb{1}_{X\ge c}\mathbb{1}_{Y\ge c}] &= \mathbb{E}[(X-Y)\mathbb{1}_{X\ge c}] - \underbrace{\mathbb{E}[(X-Y)\mathbb{1}_{X\ge c}\mathbb{1}_{Y<c}]}_{\ge 0} \\ &\le 0\end{aligned}$$

By analogous arguments, $\mathbb{E}[(Y-X)\mathbb{1}_{X\ge c}\mathbb{1}_{Y\ge c}] \le 0$. Thus $\mathbb{E}[(X-Y)\mathbb{1}_{X\ge c}\mathbb{1}_{Y\ge c}] = 0$. Running it back, this implies $\mathbb{E}[(X-Y)\mathbb{1}_{X\ge c}\mathbb{1}_{Y<c}] = 0$ for all c. I claim that this implies $\mathbb{P}(X \ge c \text{ and } Y < c) = 0$ for all c. Indeed, suppose this has positive measure; then it is the countable union of $\mathbb{P}(X \ge c \text{and} Y < c - \frac{1}{n})$, thus one of these has positive measure; but then

$$\mathbb{E}[(X-Y)\mathbb{1}_{X\ge c}\mathbb{1}_{Y<c-\frac{1}{n}}] \ge \frac{1}{n}\mathbb{P}(X \ge c \text{ and } Y < c - \frac{1}{n}) > 0$$

contradiction. Okay! Thus $\mathbb{P}(X \ge c \text{ and } Y < c) = 0$ for all c. Taking the union across all rational c, $\mathbb{P}(X > Y) = 0$. By symmetry, $\mathbb{P}(X < Y) = 0$. Thus $X = Y$ almost surely.

Solution 8

Problem 8 — Let X, Y be two independent Bernoulli random variables with parameter $p \in (0,1)$. Let $Z = \mathbb{1}_{X+Y=0}$. Compute $\mathbb{E}[X|Z]$ and $\mathbb{E}[Y|Z]$.

You can just compute everything. $\mathbb{P}(X=0, Y=0) = (1-p)^2$, $\mathbb{P}(X=1, Y=0) = p(1-p)$, $\mathbb{P}(X=0, Y=1) = p(1-p)$, and $\mathbb{P}(X=1, Y=1) = p^2$. Thus,

$$\mathbb{E}[X|Z] = \begin{cases} \frac{1}{2-p} & \text{if } Z = 0 \\ 0 & \text{if } Z = 1 \end{cases}$$

and ditto for Y.

Solution 9

Problem 9 — Give an example of a random variable X and two σ-algebras $\mathfrak{H}$ and $\mathfrak{G}$ such that X is independent of $\mathfrak{H}$ and $\mathfrak{G}$ is independent of $\mathfrak{H}$ but nevertheless $\mathbb{E}[X|\sigma(\mathfrak{H},\mathfrak{G})] \neq \mathbb{E}[X|\mathfrak{G}]$.

Let Y, Z be two independent fair coin flips and let $\mathfrak{G}$ and $\mathfrak{H}$ be their sigma algebras. Then, $X = \mathbb{1}_{Y+Z=1}$ is independent of $\mathfrak{G}$ and $\mathfrak{H}$, but $\mathbb{E}[X|\sigma(\mathfrak{G},\mathfrak{H})] = X$ while $\mathbb{E}[X|\mathfrak{G}] = \frac{1}{2}$.